Chicago Housing Data

Scope

This folder contains property sales, property characteristics, ZIP lookups, and location and economic measures for the Chicago housing project. The sales file contains 77,493 transactions for 63,701 properties, dated September 1, 2019 through August 11, 2026. The filenames describe the requested extraction window; they do not guarantee observations through September 2026.

Field definitions were checked on October 13, 2026 against the original publishers' field dictionaries and ACS table metadata. Row counts and missingness describe the local files. Derived columns and extraction choices are distinguished from original source fields; unknown missing-value conventions remain unresolved.

Sources and Verification Status

Dataset	Original source	Verification status
Sales	Cook County Parcel Sales · field dictionary	Local field names match the source API fields. The extraction query and filters were not supplied.
Characteristics	Cook County Improvement Characteristics · field dictionary	Definitions verified. Original source is building-level; how the local file became unique by property–year remains unconfirmed.
ZIP lookup	Cook County Parcel Universe · field dictionary	Compatible field definitions; this is a likely source, not confirmed extraction provenance.
Crime	Chicago Crimes: 2001 to Present · field dictionary	Likely incident source. ZIP assignment, count construction, and violent/property classifications require the preparation code.

Dataset	Original source	Verification status
Mortgage	FRED MORTGAGE30US	Original is weekly, percent, not seasonally adjusted. Local workbook is a monthly export; confirm averaging settings.
Tax	ACS B25103 and B25077	Source concepts checked; tax flag and ratio are project-derived.
Migration	ACS B07003 and B01003	Source concepts checked; shares are project-derived.
ZIP unemployment	ACS B23025	Civilian labor force and unemployed concepts checked; rate is project-derived.
County unemployment	FRED ILCOOK1URN	BLS monthly unemployment rate, percent, not seasonally adjusted.

The four ACS-based CSVs were obtained through Census Reporter according to the supplied provenance. They represent **ZCTAs**, not exact postal ZIP boundaries, and five-year estimates, not annual 2024 observations. Source variable codes below document the intended mapping, not a row-by-row reconciliation with a fresh API download. Local ratio formulas were verified across every row to four decimal places.

`zip_summary_for_tableau.csv` is a derived visualization output, not a tenth original dataset. It was referenced in Drive but is not present in the local data folder; its fields have not been checked.

Joining the Files

Keep `pin` and `zip_code` as strings to preserve leading zeros. Use sales as the transaction-level base table.

File	Join key	Grain and handling
Property characteristics	`pin`, `year`	One row per property and year in this supplied file; validate a many-to-one join.

File	Join key	Grain and handling
ZIP lookup	`pin`, `year`	One row per property and year; some ZIP values are blank.
Crime	`zip_code`, prior sale year	Annual ZIP-level counts; a 2026 sale uses 2025 crime.
Tax, migration, ZIP unemployment	`zip_code`	One ACS 2020–2024 snapshot per ZIP.
Mortgage rate	Sale year and month	Monthly rate; read the `Monthly` worksheet and locate the actual header row.
County unemployment	Sale year and month	Monthly county-level rate.

Check that the transaction count stays unchanged after each left join. Check unmatched records and missing values separately: a matched row can still contain a blank value.

Field Definitions

Sales

File: `sales_chicago_2019-09_2026-09.csv` · 77,493 rows · 9 fields

Field	Definition	Type/Unit	Preparation notes
`pin`	Parcel Identification Number (PIN).	Identifier / 14-digit string	Use for joins and grouping repeat sales; preserve leading zeros.
`year`	Year field in the sales source.	Year	Source dictionary says only Year; derive transaction year from `sale_date` and check consistency.
`township_code`	Assessor township code.	Category	Numeric codes are labels, not quantities.
`nbhd`	Five-digit Assessor neighborhood code: first two	Category / string	The full code already includes township; it is not a named Chicago community area.

Field	Definition	Type/Unit	Preparation notes
	digits identify township, last three identify neighborhood.		
`class`	Assessor property classification.	Category	Class 234 denotes split-level residences with a lower level below grade, all ages and sizes; verify year-appropriate definitions when grouping other codes.
`sale_date`	Sale recording date, not execution/closing date, according to the source dictionary.	Date	Source also documents IDOR date corrections. The omitted `is_mydec_date` flag prevents checking which dates were overwritten.
`sale_price`	Recorded transaction price.	USD	Regression target; investigate unusual transactions before filtering.
`deed_type`	Recorded deed category.	Category	Observed values include Warranty, Trustee, and Other. A category alone does not establish whether a sale is at market value.
`sale_type`	Recorded sale description.	Category	Mostly blank; populated values in this file are LAND AND BUILDING.

Source sales may include non-market transactions. The publisher's October 2023 update says filtering is no longer automatic, although an older filtering note remains on the page. This local extract omits `sale_filter_same_sale_within_365`, `sale_filter_less_than_10k`, `sale_filter_deed_type`, `is_multisale`, `num_parcels_sale`, and sale document numbers. Confirm extraction filters and multi-parcel handling with the preparer; a parcel-row price cannot automatically be assumed to represent only that parcel.

Property Characteristics

File: `chars_chicago_2019-09_2026-09.csv` · 142,617 rows · 23 fields

Field	Definition	Type/Unit	Preparation notes
pin	Property identification number.	Identifier / string	Join with sale pin and year.
year	Tax year associated with the characteristics record.	Year	Tax-year data need not describe the property precisely as of its sale date.
card	Improvement/building number within a parcel.	Identifier	Original source can have multiple buildings per property–year. This local extract is unique by property–year; confirm selection/aggregation before treating area as parcel-total area.
class	Assessor property classification in the characteristics record.	Category	May differ from the sale record; reconcile before selecting a source.
township_code	Assessor township code.	Category	May duplicate the sales field after merging.
char_yrblt	Recorded year built.	Year	Can derive home age as sale year minus year built.
char_bldg_sf	Building area measured from the exterior.	Square feet	Label as building area, not living area; investigate nonpositive values.
char_land_sf	Recorded land/lot area.	Square feet	Examine skewness and possible log transformation.
char_beds	Recorded number of bedrooms.	Count	Check overlap with rooms and building area.
char_rooms	Total rooms in the building, excluding bathrooms.	Count	Includes more than bedrooms; check overlap with bedrooms and building area.
char_fbath	Bathrooms with a bathtub or shower.	Count	Source documentation specifies a default of 1 when missing; the local extract still has blanks. Document any decision to apply

Field	Definition	Type/Unit	Preparation notes
			that source default rather than mean imputation.
char_hbath	Bathrooms without a bathtub or shower.	Count	Do not assume a blank means zero without source confirmation.
char_bsmt	Basement type/category.	Category	Examples include Full.
char_bsmt_fin	Basement finish category.	Category	Examples include Unfinished; distinguish absent basement from unknown finish.
char_gar1_size	Recorded size/capacity category of the first garage.	Text category / car capacity	Values such as 0 cars and 1.5 cars may be parsed after checking all categories.
char_air	Central air-conditioning category.	Category	Does not measure all forms of air conditioning.
char_ext_wall	Exterior wall material/category.	Category	Examples include Masonry.
char_use	Single-family versus multi-family use.	Category	The source defines this narrower use distinction.
char_type_resd	Residential structure type/style.	Category	Examples include 1.5 Story.
char_renovation	Recorded renovation category.	Category	Source labels it Renovation without a timeframe or missing-value rule. Do not equate it with the separate recent-renovation model feature. Approximately 99% blank locally.
char_repair_cnd	Recorded repair/condition rating.	Category	Mostly Average; other observed categories include Above Average and Below Average.
char_cnst_qlty	Recorded construction-quality rating.	Category	Mostly Average; other observed categories include Deluxe and Poor.

Field	Definition	Type/Unit	Preparation notes
`tieback_proration_rate`	Share of assessed value allocated to this PIN when value is divided across tied-back PINs.	Proportion	Used to allocate taxable assessed value; distinct from card-level proration. It is not automatically a multiplier for sale price or building area.

The publisher notes a characteristics-history change: records before 2021 omit Home Improvement Exemption characteristic updates, whereas records from 2021 include them. Current-year records can also be incomplete before assessment-roll certification. Source documentation. Class 234's description is also shown on an official Assessor property record.

ZIP Lookup

File: `zip_chicago_2019-09_2026-09.csv` · 74,710 rows · 3 fields

Field	Definition	Type/Unit	Preparation notes
`pin`	Property identification number.	Identifier / string	Join on property and year.
`year`	Tax year, if Parcel Universe is the extraction source.	Year	Unique with `pin` locally; provenance remains unconfirmed.
`zip_code`	Property ZIP code, if Parcel Universe is the extraction source.	Category / string	Preserve leading zeros; do not scale as a numeric quantity. This is not a Census ZCTA field.

Crime

File: `crime_by_zip_2018_2025.csv` · 472 rows · 59 ZIPs × 8 years

Field	Definition	Type/Unit	Preparation notes
`zip_code`	Project-assigned ZIP/ZCTA grouping for crime records.	Category / string	Original incident source has no ZIP field; confirm spatial boundaries, coordinate handling, and unmatched records.

Field	Definition	Type/Unit	Preparation notes
`year`	Intended year of incident occurrence.	Year	Original source defines year by occurrence, not reporting date. Confirm the aggregation used this field; use prior-year values for sales.
`crimes_total`	Project-derived count of included source records by ZIP and year.	Count	Confirm whether records or distinct incident identifiers were counted. Source has a homicide victim-level exception.
`crimes_violent`	Project-derived count classified as violent.	Count	No matching original aggregate field; obtain exact IUCR, FBI, or primary-type classification rules.
`crimes_property`	Project-derived count classified as property crime.	Count	No matching original aggregate field; obtain exact classification rules.

The original source records reported incidents, except homicide records can represent individual victims. Its location coordinates are redacted to the block level; records and classifications can be revised. Counts do not measure all crime that occurred. The publisher cautions against comparisons over time; disclose reporting and classification limitations in trend analysis. Source dictionary.

Population-normalized rates can be calculated using `pop_total`, but this applies a fixed ACS population snapshot to multiple years. Downtown visitor activity and ZIPs crossing city boundaries limit comparability.

Mortgage Rate

File: `MORTGAGE30US.xlsx` · worksheets `README` and `Monthly`

Field	Definition	Type/Unit	Preparation notes
`observation_date`	Month associated with the rate observation.	Date	The actual field names appear below the first worksheet row; inspect the header before loading.

Field	Definition	Type/Unit	Preparation notes
MORTGAGE30US	Freddie Mac national 30-year fixed mortgage rate, exported at monthly frequency locally.	Percent, not seasonally adjusted	Original FRED series is weekly, ending Thursday. Monthly averaging is the reported project transformation; workbook confirms monthly frequency but not the aggregation setting.

FRED documents a survey-method change on November 17, 2022. For prediction at a particular date, a completed same-month average would not yet be available.

ZIP Tax Measures

File: tax_by_zip_acs2020_2024.csv · 58 rows

Field	Definition	Type/Unit	Preparation notes
zip_code	ZIP/ZCTA grouping.	Category / string	ACS geography approximates postal ZIPs.
acs_period	ACS estimate period.	Period label	One 2020–2024 snapshot.
median_re_tax_paid	Median annual real-estate taxes paid for owner-occupied housing units; intended source B25103_001E.	USD / year	Five local values are 10001 with tax_topcoded=1; treat them as flagged/censored values, not exact $10,001 bills. Confirm the original annotation and preparer's encoding.
tax_topcoded	Project-created indicator for a censored/top-coded tax estimate.	Binary: 0/1	Not an original B25103 estimate field; locally matches values of 10001.
median_home_value	Median value of owner-occupied housing units; intended source B25077_001E.	USD	Neighborhood estimate, not a transaction price. Excluding it is a model-design choice; historical availability still requires checking.
eff_tax_rate_proxy	Median tax paid divided by median home value.	Proportion	0.0162 means 1.62%; this is a ratio of medians, not an individual property tax rate. Five

Field	Definition	Type/Unit	Preparation notes
			values are missing; do not fill with zero.

ZIP Migration Measures

File: `migration_by_zip_acs2020_2024.csv` · 58 rows

Field	Definition	Type/Unit	Preparation notes
`zip_code`	ZIP/ZCTA grouping.	Category / string	Match geography carefully.
`acs_period`	ACS estimate period.	Period label	One 2020–2024 snapshot.
`pop_total`	Estimated total population; intended source `B01003_001E`.	People	Available denominator for approximate crime rates.
`pop_1yr_plus`	Estimated residents aged one year and older; intended source `B07003_001E`.	People	Denominator for migration shares.
`same_house`	Residents aged one or older in the same house one year earlier; intended source `B07003_004E`.	People	Residence comparison at survey time across the five-year estimation period.
`moved_within_county`	Residents aged one or older who moved within the same county; intended source `B07003_007E`.	People	Excluded from outside-county in-migration.
`moved_from_other_county_IL`	Residents aged one or older who moved from a different county in the same state; intended source `B07003_010E`.	People	IL is valid because the destination ZCTAs are in Illinois; source label is same state.
`moved_from_other_state`	Residents aged one or older who moved from another state; intended source `B07003_013E`.	People	Component of outside-county in-migration.
`moved_from_abroad`	Residents aged one or older who moved from abroad; intended source `B07003_016E`.	People	Geographic origin, not citizenship or immigration status.

Field	Definition	Type/Unit	Preparation notes
in_migration_share	Share of residents aged one or older who moved from outside the county.	Proportion	Sum of the three outside-county origin counts divided by pop_1yr_plus; verify supplied rounding.
mover_share	Share of residents aged one or older who changed residence.	Proportion	Includes within-county moves; approximately 1 minus same_house / pop_1yr_plus.

These are five-year survey estimates of one-year residence status, not a count of moves during the entire 2020–2024 period. in_migration_share is not net migration and does not subtract people who left. Same-county movement can cross ZIP boundaries, so this share is specifically outside-county migration, not all moves into a ZIP. The five residence-status counts sum to pop_1yr_plus in every local row; both shares match their formulas to four decimal places.

ZIP Unemployment

File: unemployment_by_zip_acs2020_2024.csv · 58 rows

Field	Definition	Type/Unit	Preparation notes
zip_code	ZIP/ZCTA grouping.	Category / string	Match to sale ZIP.
acs_period	ACS estimate period.	Period label	One 2020–2024 snapshot.
civ_labor_force	Estimated civilian labor force aged 16 and older; intended source B23025_003E.	People	Includes employed and unemployed civilians, not everyone aged 16 and older.
unemployed	Estimated unemployed civilians aged 16 and older in the labor force; intended source B23025_005E.	People	Numerator for unemployment rate.
unemployment_rate	Unemployed divided by civilian labor force.	Proportion	0.0335 means 3.35%; differs in scale from the county monthly file.

County Monthly Unemployment

File: `unemployment_cook_county_monthly_2018_2026.csv` · 104 rows

Field	Definition	Type/Unit	Preparation notes
`date`	Month associated with the unemployment observation.	Date	Join by sale year and month.
`cook_unemployment_rate`	BLS Cook County unemployment rate from FRED `ILCOOK1URN`: unemployed as a percentage of the civilian labor force.	Percent, monthly, not seasonally adjusted	5.0 means 5%; one local missing value. Publication lags and revisions matter for historical prediction.

Preparation and Modeling Notes

- Fit imputation, scaling, and categorical encoding only on training data. Group repeat sales by `pin` when splitting.
- Fixed ACS ZIP measures are redundant with a complete set of ZIP indicators. Compare location-category and location-characteristic model specifications separately.
- Full year–month indicators absorb mortgage and county unemployment measures that vary only by month.
- ACS 2020–2024 five-year estimates were released on **January 29, 2026**, following a revised schedule. The previous December 2025 statement was incorrect. Disclose retrospective use for earlier sales. Census release announcement.
- Treat 2019 and 2026 as partial sale years. Check actual date coverage before comparing annual totals.
- Missing renovation, garage, basement, and bathroom fields require semantic checks before imputation.
- The supplied market excludes condos according to the original review. HOA fees are unavailable, and ZIP/ZCTA measures do not describe individual households.
- ACS margins of error and source annotations are absent from these CSVs. Small-population ZCTA estimates can be uncertain; the local files do not support direct uncertainty assessment.